

Solid State Physics

Past Paper fall 2021

Question # 1:-

Give short answers of the following.

— (a) —

Diff b/w primitive and non-primitive with the help of diagram.

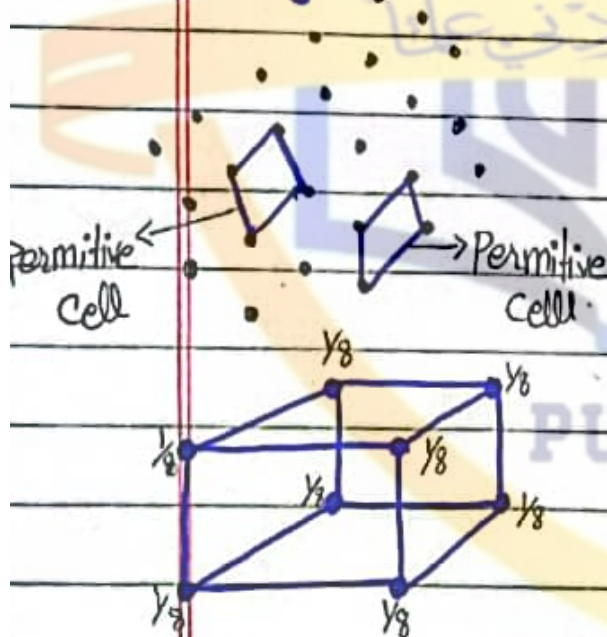
Primitive	non-Primitive:
Primitive unit Cell maybe define as a cell geometrical shape, which when repeated indefinitely in three dimension will fill all shape without any overlapping or leaving gap and	Non-primitive unit Cell contain additional lattice points, either on a face of the unit Cell or with in the unit Cell and So have more than one

equivalent of one lattice point per point only at the corner unit Cell. in known as primitive Cell.

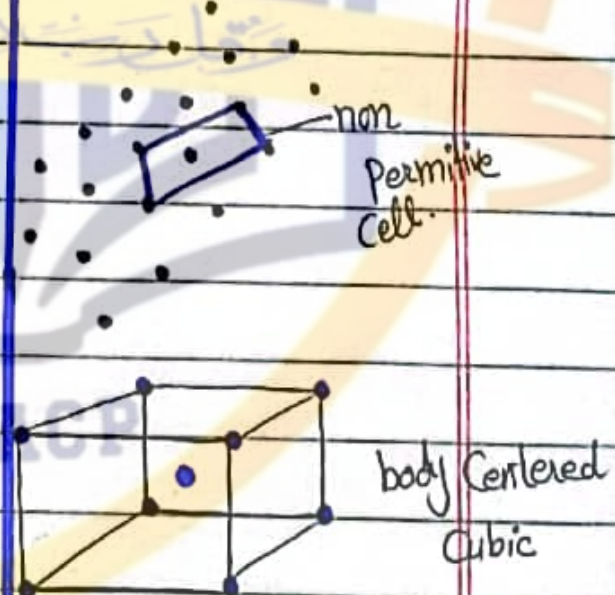
Example:-

Diamond and any Diamond like materials (Silicon, Graphs etc).

Diagram:-



Face Centered Cubic and body Centered Cubic are the examples of non-primitive Cell.



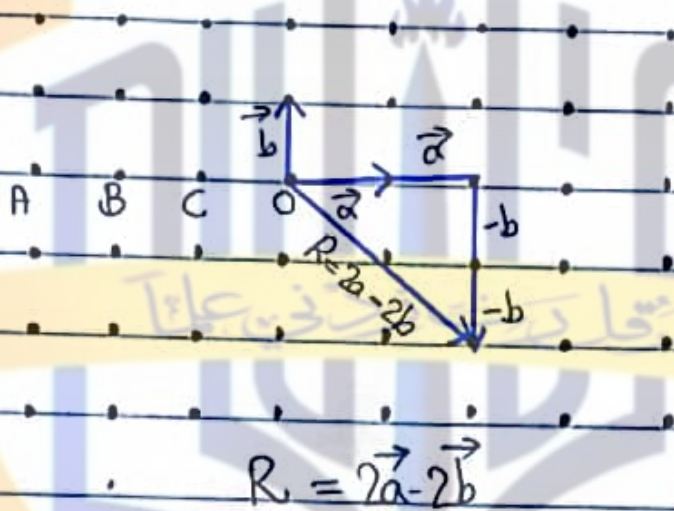
$$\text{Lattice Point} = \frac{1}{8} \times 8 + 1$$

$$= 2 \text{ per Cell.}$$

(b)

Define Bravais and non-bravais lattice with the help of diagram.
Bravais Lattice-

The most important feature about Bravais lattice is that translation symmetry hold in it.



$$R = 2\vec{a} - 2\vec{b}$$

If two points cannot be differentiate then translation Symmetry holds. Such arrangement in which

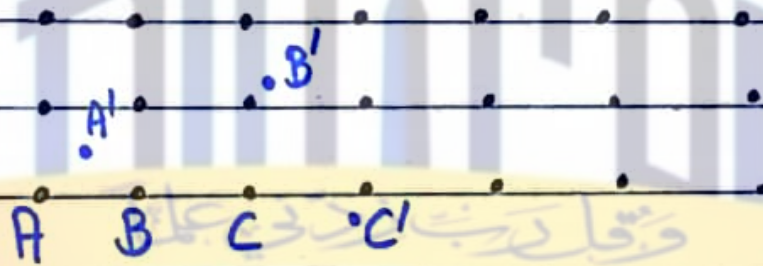
- ⇒ All points are identical.
- ⇒ Any point can be chosen as a reference point.
- ⇒ Environment around any point remain same as any other point.

All above conditions holds that arrangement holds translation Symmetry.

Non-Bravais lattice:-

In a non Bravais lattice some of the points are non-equivalent and in addition of the atoms may or may not be of same kind.

If atoms are of same kind but some of the points are non equivalent then the lattice is non-Bravais.



If atoms are of different kinds but the points are all at equidistance from each other even then lattice is said to be non-bravais.



(c)

Write down the importance of Miller indices for indexes the atomic planes.

A notation system in Crystallographic to describe different planes and directions in Bravais lattice.

It was introduced by British mineralogist W.H. Miller in 1839. That planes are called Millerian planes and the directions are called Millerian direction.

Miller indices are whole numbers enclosed in round brackets, they are used to indicate the Crystallographic planes written as:

$(h\ k\ l)$ without commas

$(h\ k\ l)$ represents the plane.

$\{h\ k\ l\}$ represents family of planes

$[U\ V\ W]$ represent the direction

$\langle U\ V\ W \rangle$ represent family of direction.

Importance:-

In material Science
It is important to have a
rotation system for atomic planes
Since these planes influence
Optical properties, Reactivity, Surface
tension and Dislocations.

(d)
Define Brillouin Zone. Explain
Importance of Brillouin Zone in
SSP.

Brillouin Zone:-

The Wigner Seitz
Cell in the reciprocal lattice
is known as Brillouin Zone.

Importance:-

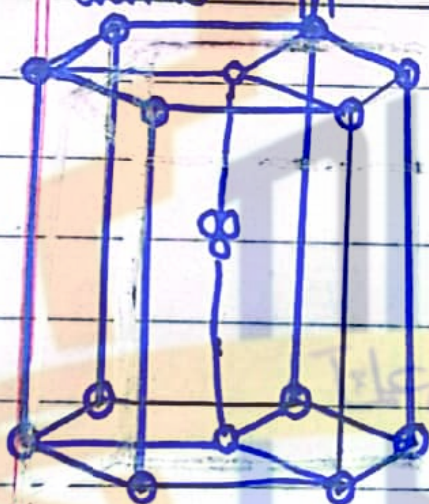
In Brillouin Zone,
all the vectors lying in that area
[2D] or Volume (in 3D) satisfy
the Bragg's Condition. The
Brillouin Zone gives a clear
geometrical interpretation of

diffraction Condition.

$$2\vec{K} \cdot \vec{G} = |\vec{G}|^2 \text{ or } \vec{K} \cdot \left(\frac{1}{2}\vec{G}\right) = \left|\frac{1}{2}\vec{G}\right|^2$$

(e)

Draw hexagonal Closed packed (hcp) structure and show all atoms in hcp unit Cell.



Hexagonal prism in the figure have 12 corner atoms, 2 face centered atoms and 3 atoms in side the body.

So, number of atoms in prism.

$$= 12 \times \frac{1}{6} + 2 \times \frac{1}{2} + 3 \times 1 = 6$$

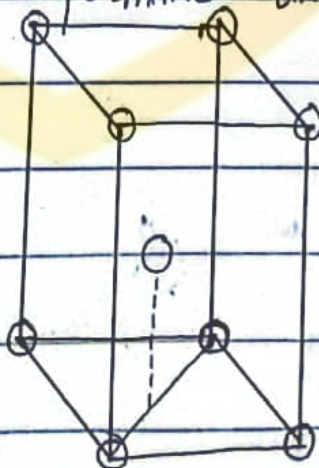
⊛ However hexagonal primitive unit Cell is.

$\frac{1}{3}$ of prism

$$a = b \neq c$$

$$\alpha = \beta = 90^\circ$$

$$\gamma = 120^\circ$$



$$6 \times \frac{1}{3} = 2$$

So Actual number of atoms in hcp Cell is 2.

(f)

Write down Bragg's equation and write what each term appearing in it represents. What is the limitation on Bragg's law to observe diffraction in Crystals?

The mathematical form of Bragg's law is given as:-

$$2d \sin \theta = n\lambda \rightarrow \text{①}$$

where λ represents the wavelength of light (X-rays), and " d " be interplanar spaces.

from eq ①

$$2d \sin \theta = n\lambda$$

$$\sin \theta = \frac{n\lambda}{2d}$$

$$(\sin \theta)_{\max} = 1 \geq \frac{n\lambda}{2d}$$

$$\Rightarrow \lambda \geq 2d$$

This indicates that " λ " must not be greater than twice the interplanar distance otherwise no

diffraction will occur. Diffraction occurs only if Wavelength of light is of the order of one Angstrom ($1\text{\AA} = 10^{-10}\text{m}$) and comparable to the size of slit.

Bragg's Limitations:-

The Bragg's law does not only mean the constructive interference of Bragg's equation, but it has

"Diffraction will never occur if Bragg's equation is not satisfied, in case of infinitely large number of lattice planes"

(9)

Define the term:- (i) Cohesive energy (ii) Madelung energy. Explain the difference in these two energies.

Cohesive energy:-

The energy required to separate the atoms to

an infinite distance apart. generally it is of the order of few eV (electron Volt)

• Cohesive energy deals with the overall stability of a Material.

Madlung Energy:-

The long range interaction between ions with charge $\pm q$ is the electrostatic interaction $\pm q^2/r$ attractive between ions of opposite charge and repulsive between ions of same charges.

The ions arrange themselves in whatever Crystal structure gives the strongest attractive interaction compatible with the repulsive interaction at short distances between ion cores. The repulsive interaction between ions with inert gas configuration are similar to those between inert gas atoms. The Van der Waals part of the attractive interactions in ionic crystal makes a relatively small contribution

to the Cohesive energy in ionic Crystals of the order 1 or 2 percent.

The main Contribution to the binding energy of ionic Crystals is electrostatic and is called the **Modelung energy**.

⇒ Modelung energy Specifically addresses the electrostatic interactions within ionic Solids.

(h)

Explain the terms group and phase Velocity. Which one is responsible for transmission in one medium?

Phase Velocity:-

"The rate at which the phase the Wave propagation in Space is Called Phase Velocity."

⇒ Represented by

$$V_p = \frac{\lambda}{T}$$

⇒ In terms of angular frequency

it can be expressed as:-

$$V_p = \frac{\omega}{K}$$

where "K" is called wave number.

$$K = \frac{2\pi}{\lambda}$$

Group Velocity:-

The velocity of the wave packet is called group velocity.

⇒ It is given as:-

$$V_g = \frac{d\omega}{dK}$$

where "K" is angular wave number

⇒ The energy transport velocity is defined as: the ratio of power flow density to stored energy density for an extended uniform transmission medium

(ii)

Define photon and phonon.
What is the characteristic of

photon and phonon?

Phonon:-

"The lattice vibrations generates elastic waves in crystal. The energy of such a wave is said to be quantized or consideration/considered to be quantized and is called phonon."

Photon:-

A photon is a fundamental particle that represents the smallest unit of light or electromagnetic radiation. It has zero rest mass and carries a quantum of energy proportional to its frequency.

Characteristic of phonon:-

- ⇒ The phonon has an analogy with the photon which is the quantum of electromagnetic energy. Thus all kind of lattice vibrations are composed of phonon.
- ⇒ The phonon is the quantum of sound, usually in a solid.

Characteristic of Photon:-

⇒ Photon behave both as a particles and wave. They travel at the Speed of light in a Vacuum and Can interact with matter, Causing effects Such as, absorption, emission and Scattering.

⇒ The photon is the quantum of electromagnetic radiation.

(j)

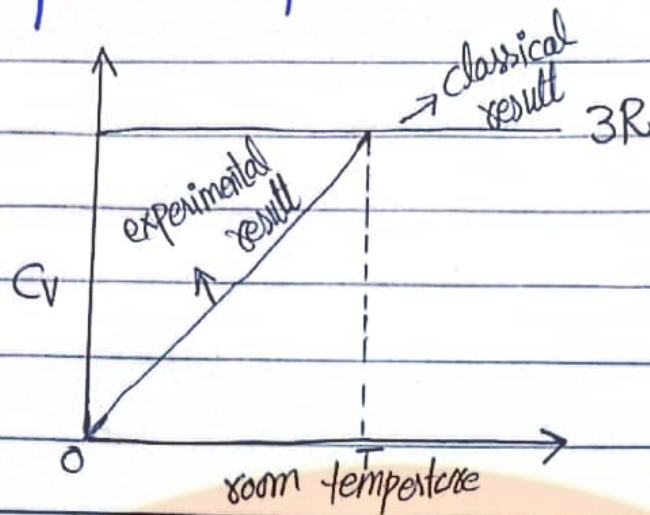
Define Dulong - Petit law of heat Capacity? Draw a heat Capacity Versus temperature plot to Support Statement.

$$C_v = 3R$$

It is classical result which says Specific heat is Constant ($3R$) at all temperatures.

It is also know as Dulong and petit law:-

Graphical Explanation:-



It is clear from the graph, that classical prediction is that C_v is constant at all temperature. But experimental results are showing that at room temperature or above the room temperature is constant but below the room temperature is **Zero** which classical theory cannot able to explain.

It is big failure of classical theory to explain specific heat at low temperature. Einstein and Debye explain this phenomena.

~~$\alpha(k)$~~

What are anharmonic Crystals interactions? Explain briefly.

Anharmonic Crystal Interaction:-

The harmonic theory used in the lattice vibrations of solids assumes that the anharmonic terms in the lattice potential energy expansion are neglected while the quadratic term is retained.

This assumption may have several consequences.

- 1 There is no thermal expansion
- 2 Interaction between lattice waves does not exist and single elastic wave does not decay or change its form with time.
- 3 Elastic constants are equal in both adiabatic and isothermal process.
- 4 Elastic constants do not depend on pressure and temperature.
- 5 The heat capacity becomes constant at high temperature when $T \gg \theta$

What is meant by quantization of elastic waves? Discuss briefly.

Quantization of Elastic Waves:-

The energy of a lattice vibrations is quantized. The quantum of energy is called a phonon. The energy of an elastic mode of an angular frequency ω is

$$\epsilon = (n + \frac{1}{2}) \hbar \omega$$

Consider the standing wave of amplitude

$$u = u_0 \cos kx \cos \omega t$$

Here u is the displacement of a volume element from its equilibrium position at x in the crystal. The

Kinetic energy is $\frac{1}{2} \rho \left(\frac{\partial u}{\partial t} \right)^2$ where ρ is the mass density. In a crystal of volume V , the volume integral of the kinetic energy is $\frac{1}{4} \rho V \omega^2 u_0^2 \sin^2 \omega t$

The time average kinetic energy is

$$\frac{1}{8} \rho V \omega^2 u_0^2 = \frac{1}{2} (n + \frac{1}{2}) \hbar \omega$$

because $\langle \sin^2 \omega t \rangle = \frac{1}{2}$ The square of the amplitude of the 2 mode is:-

$$U_0^2 = 4(n + \frac{1}{2}) \hbar / \rho \omega V$$

This relates the displacement in a given mode to the phonon occupancy n of the mode.

Q(m)

What is the Origin of Van der Waals interaction in Crystals? What is its nature and how it varies between two identical inert gas atoms?

Van der Waals Interactions:-

"When adjacent atoms come close enough that their electron clouds just barely touch. This action induces charge fluctuations that result in a non-specific, non-directional attraction called Van der Waals Interactions."

→ The origin of Van der Waals interaction in the Van der Waals

bonds. It is a dipole-induced dipole interaction which exists in two neighbouring atoms so as to produce weak attractive force between atoms. These interaction are also known as dispersion bonds.

Nature:-

- (i) Polarizability
- (ii) The strength of Van der Waals forces decreases rapidly with distance.

Two Identical Inert gas atoms:-

Van der Waals forces, Specifically London dispersion forces, between two identical inert gas atoms

Helium

Neon

Argon

as primarily influenced by their polarizability and the distance between them.

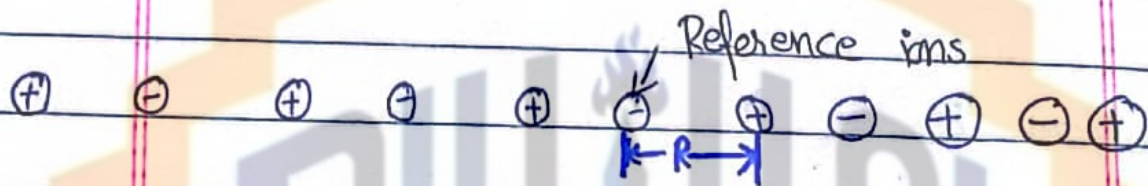
(n)

Calculate the Value of Modeling

Constant for a linear chain of alternative positive and negative ions separated by a repeat distance R .

$$\alpha = \sum_j \pm \frac{1}{P_{ij}}$$

A line of ions of alternating sign with distance b/w ions is R .



By dividing R on bs of eq ①

$$\frac{\alpha}{R} = \sum_j \pm \frac{1}{P_{ij} R}$$

$$\frac{\alpha}{R} = \sum_j \pm \frac{1}{R_{ij}}$$

$$\frac{\alpha}{R} = \left[\frac{1}{R} - \frac{1}{2R} + \frac{1}{3R} - \frac{1}{4R} + \dots + \frac{1}{R} - \frac{1}{2R} + \frac{1}{3R} - \frac{1}{4R} + \dots \right]$$

$$\frac{\alpha}{R} = \left[\frac{2}{R} - \frac{2}{2R} + \frac{2}{3R} - \frac{2}{4R} + \dots \right]$$

$$\frac{\alpha}{R} = \frac{2}{R} \left[1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \right]$$

$$\frac{\alpha}{R} = \frac{2}{R} \ln(1+1) \Rightarrow \frac{\alpha}{R} = \frac{2}{R} \ln 2$$

$$\alpha = 2 \ln 2$$

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$$\alpha = 2 \ln 2$$

$$\alpha = 1.386$$

$$\therefore \ln(1+n) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$\text{If } n=1$$

$$\therefore \ln(1+1) = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

If is value of Madelung Constant:

Value of Madelung Constant for different Crystals are

$$\text{NaCl} = \alpha = 1.7475$$

$$\text{CsCl} = \alpha = 1.7626$$

$$\text{ZnS} = \alpha = 1.6381$$

~~(0)~~

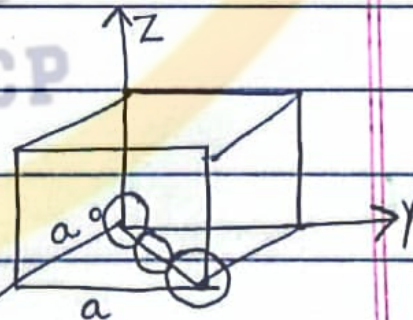
Calculate the linear density of an FCC Crystal along $[110]$.

Formula for linear density =

$$\text{Linear density} = \frac{\text{No of atoms on line}}{\text{length of line}}$$

Total no of atoms

$$\begin{aligned} \text{axe} &= \frac{1}{2} + 1 + \frac{1}{2} \\ &= 2 \end{aligned}$$



length of line is x

$$c = \sqrt{a^2 + a^2}$$

$$c = \sqrt{2}a^2$$

$$c = \sqrt{2}a$$

By pythagoras theorem

$$c^2 = a^2 + a^2$$

$$c = \sqrt{a^2 + a^2}$$

linear density of FCC at $[110]$ is:-

$$L.D_{[110]} = \frac{2}{\sqrt{2}a} \Rightarrow \boxed{\sqrt{2}a = L.D_{[110]}} \rightarrow \textcircled{1}$$

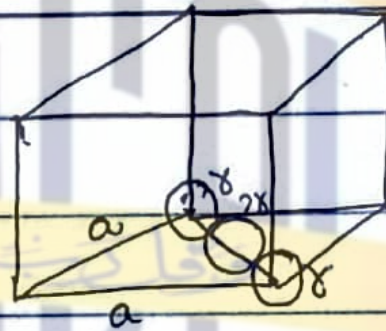
For Value of a :-

$$a = \frac{4x}{\sqrt{2}}$$

Putting this value in
equation $\textcircled{1}$

$$L.D_{[110]} = \frac{\sqrt{2}}{4x} \cdot \sqrt{2}$$

$$L.D_{[110]} = \frac{1}{2x}$$



By Pythagoras
 $(4x)^2 = a^2 + a^2$

$$x = \frac{\sqrt{2}a}{4}$$

$$a = \frac{4x}{\sqrt{2}}$$

LONG Q (A)

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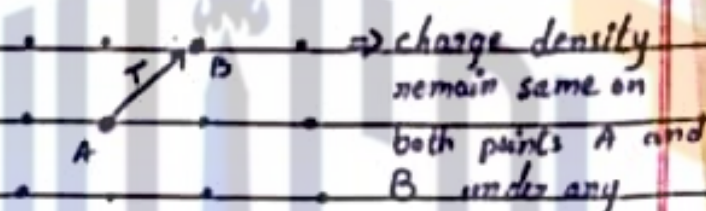
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Fourier Analysis

Fourier series is an infinite series representation of periodic function in terms of trigonometric function $\sin\theta$ and $\cos\theta$.

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

2D lattice:



Thus

translation $\vec{T}$

$n(\vec{r} + \vec{T}) = n(\vec{r}) \Rightarrow$ such periodicity creates an ideal situation for F.A

Consider function $n(x)$ in one dimension and expand

with Fourier series:

$$n(x) = n_0 + \sum_{p=1}^{\infty} [C_p \cos(2\pi p x/a) + S_p \sin(2\pi p x/a)] \quad \dots (1)$$

where $p =$ positive integer, $C_p, S_p =$ Fourier coefficients

$\rightarrow$ factor $2\pi/a$ ensures that $n(x)$ has period a .

$$n(x+a) = n_0 + \sum_{p=1}^{\infty} [C_p \cos(2\pi p(x+a)/a) + S_p \sin(2\pi p(x+a)/a)]$$

$$n(x+a) = n_0 + \sum_{p=1}^{\infty} [C_p \cos(2\pi p x/a + 2\pi p) + S_p \sin(2\pi p x/a + 2\pi p)]$$

$$= n_0 + \sum_{p=1}^{\infty} [C_p \cos 2\pi p x/a + S_p \sin(2\pi p x/a)]$$

$$n(x+a) = n(x)$$

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$\Rightarrow 2\pi p/a$ point in the reciprocal lattice

$\Rightarrow$ In 1D these points lie on the line

$$\begin{array}{ccccccc} & n & & & & & \\ & \cdot & & \cdot & & \cdot & \\ x & x & x & x & x & x & x \end{array}$$

$\Rightarrow$ tells Allowed elements in the Fourier series

$$n(x) = \sum_p n_p \exp(i2\pi p x/a) \quad (5)$$

$p =$ positive integer, $-$ int., 0

$n_p =$ complex numbers

To ensure $n(x)$ is real, we take

$$n_p^* = n_{-p} \quad (6)$$

Let $\phi = 2\pi p x/a$ put in (5)

$$n(x) = \sum_p n_p e^{i\phi}$$

$$n(x) = n_p(\cos\phi + i\sin\phi) + n_{-p}(\cos\phi - i\sin\phi) \quad \because e^{i\phi} = \cos\phi + i\sin\phi$$

$$= n_p \cos\phi + n_p i \sin\phi + n_{-p} \cos\phi - n_{-p} i \sin\phi$$

$$= (n_p + n_{-p}) \cos\phi + i(n_p - n_{-p}) \sin\phi$$

From (6)

$$= (n_p^* + n_p) \cos\phi + i(n_p^* - n_p) \sin\phi$$

$$= 2 \operatorname{Re}\{n_p\} \cos\phi + i 2 \operatorname{Im}\{n_p\} \sin\phi$$

$$n(x) = 2 \operatorname{Re}\{n_p\} \cos\phi - 2 \operatorname{Im}\{n_p\} \sin\phi$$

Real so $n(x)$ is real

So, For 3D

$$n(\vec{r}) = \sum_{\vec{G}} n_{\vec{G}} \exp(i\vec{G} \cdot \vec{r})$$

$$\begin{array}{l} n_p = (a+ib) \\ n_{-p} = (a-ib) \end{array}$$

$$\therefore (a+ib)(a-ib) = 2a$$

real part of n_p

From 6

$$2 \operatorname{Re} n_p$$

Similarly

$$n_p - n_{-p} = a+ib - a-ib$$

$$= 2ib$$

Im part

of n_p

From 6

$$2 \operatorname{Im} n_p$$

Inverse of Fourier Series

$$n_p = a^{-1} \int_0^a dx \, n(x) \exp(-i2\pi p x/a)$$

$$n_p = a^{-1} \sum_{p'} n_{p'} \int_0^a dx \, e^{i2\pi p' x/a} \cdot e^{-i2\pi p x/a} \quad \therefore n(x) = \sum_{p'} n_{p'} e^{i2\pi p' x/a}$$

$$n_p = a^{-1} \sum_{p'} n_{p'} \int_0^a dx \, e^{i2\pi x/a (p'-p)}$$

Now there two cases $p' \neq p$ & $p' = p$

When $p' \neq p$

$$\int_0^a dx \, e^{i2\pi (p'-p)x/a} = \frac{e^{i2\pi (p'-p)x/a}}{i2\pi (p'-p)/a} \Big|_0^a$$

$$= \frac{a}{i2\pi (p'-p)/a} [e^{i2\pi (p'-p)} - 1]$$

Now here factor $e^{i2\pi (p'-p)} = 1$

$$\int_0^a dx \, e^{i2\pi (p'-p)x/a} = 0 \quad \therefore \frac{e^{i2\pi (p'-p)} - 1}{i2\pi (p'-p)/a} = 0$$

So, $n_p = 0$

When $p' = p$

$$n_p = a^{-1} n_p \int_0^a dx$$

$$n_p = a^{-1} n_p \cdot a$$

$$n_p = a^{-1} n_p a$$

$$n_p = n_p$$

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$$n_x = \sum_p n_p e^{i2\pi p x/a}$$

$$n(x) = \dots n_{-p} e^{-i2\pi p x/a} + n_0 + n_1 e^{i2\pi x/a}$$

$$= \dots n_p (\cos \phi - i \sin \phi) + n_0 + n_1 (\cos \phi + i \sin \phi)$$

take neglect take

In generalised form

$$= n_p (\cos \phi - i \sin \phi) + n_p (\cos \phi + i \sin \phi)$$

How 1D mono-atomic linear lattice behaves as a low pass filter?

Vibrations in 1-Dimensional mono-atomic Lattice:-

The one-dimensional mono-atomic linear lattice is the simplest possible solid.

In order to understand the vibrations in one dimensional mono-atomic linear lattice following assumptions are made:

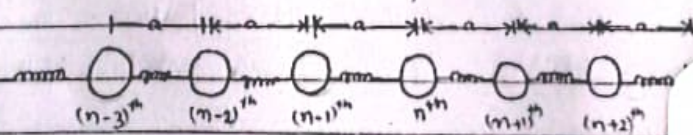
The lattice contains a very large



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- number ($N \rightarrow \infty$) of atoms.
- Each atom has same identical mass " m ".
- The intra-atomic separation between two atoms in a lattice is constant " a ".
- The atoms have short range forces i.e. only nearest neighboring atoms interact with each other.
- The interaction of an atom with its neighbor is represented by a massless elastic spring.
- The spring has spring constant " β ".
- Force in spring is proportional to the change in length. i.e. it obeys Hooke's Law.
- The generated vibrations are waves on linear chain (i.e. one-dimensional lattice) is longitudinal.

Consider a chain of atoms of one-dimensional mono-atomic lattice, as shown in figure. The separation b/w two atoms is " a " at equilibrium position.



□ state of equilibrium

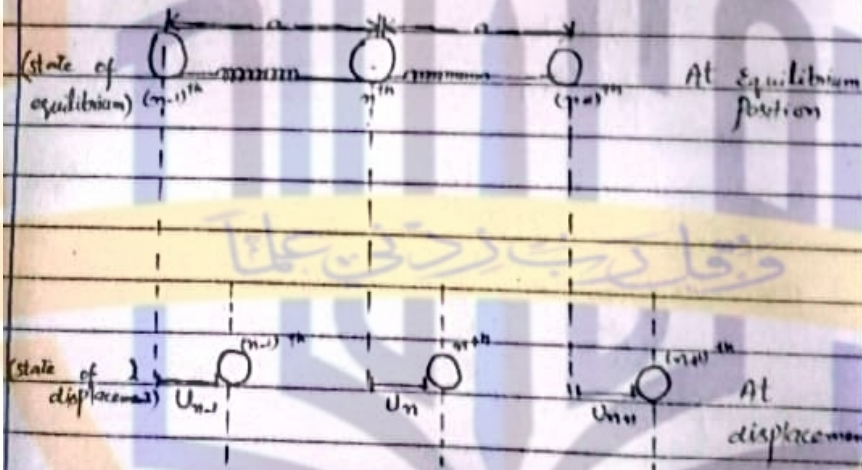


All the atoms are coupled to each other by elastic spring. Thus, the vibrations travel throughout the medium.

Let n^{th} atom as reference atom. It will interact with its neighboring $(n+1)^{\text{th}}$ and $(n-1)^{\text{th}}$ atoms.

The n^{th} atom takes displacement U_n from equilibrium position after time t when the disturbance travels. Similarly $(n+1)^{\text{th}}$ and $(n-1)^{\text{th}}$ atoms takes displacement U_{n+1} and U_{n-1} respectively after time t when disturbance travels.

The state of disturbance i.e displacement is shown as:-



Force on n^{th} atom due to $(n-1)^{\text{th}}$ atom:

$$F_1 = -\beta U$$

$$\because F = -kx$$

where β is spring constant and U is displacement by Hooke's law.

Thus:-

$$F_1 = -\beta (U_n - U_{n-1})$$

$\because U = U_n - U_{n-1}$ = effective displacement

Force on n^{th} atom due to $(n+1)^{\text{th}}$ atom:

$$F_2 = -\beta U$$

$$F_2 = -\beta (U_n - U_{n+1})$$

Net force on n^{th} atom due to its nears neighbours is given as:-

F

Thus:-

$$F_1 = -\beta(U_n - U_{n-1}) \quad \because U = U_n - U_{n-1} = \text{effective displacement}$$

Force on n^{th} atom due to $(n+1)^{\text{th}}$ atom:

$$F_2 = -\beta U$$

$$F_2 = -\beta(U_n - U_{n+1})$$

Net force on n^{th} atom due to its nearest neighbours is given as:-

$$F = F_1 + F_2$$

$$F = -\beta(U_n - U_{n-1}) - \beta(U_n - U_{n+1})$$

$$= +\beta[-(U_n - U_{n-1}) - (U_n - U_{n+1})]$$

$$= \beta[-U_n + U_{n-1} - U_n + U_{n+1}]$$

$$F = \beta[U_{n-1} - 2U_n + U_{n+1}] \rightarrow (i)$$

As:-

$$F = ma = m \frac{d^2 \vec{r}}{dt^2}$$

$$\because \vec{V} = \frac{d\vec{r}}{dt}$$

$$= m \frac{d^2 \vec{r}}{dt^2}$$

The displacement of n^{th} atom is U_n . So:-

$$F = m \frac{d^2 U_n}{dt^2} \rightarrow (ii)$$

Comparing eqⁿ (i) and (ii) -



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$$m \frac{d^2 U_n}{dt^2} = \beta[U_{n-1} - 2U_n + U_{n+1}] \rightarrow (iii)$$

This is the equation of motion.

The harmonic solution for the n^{th} atom from its mean position is:

$$U_n = U_0 e^{i(\omega t - kna)} \rightarrow (iv)$$

where U_0 = amplitude of the vibration and it will be assumed that all the atoms have same amplitude.

As the displacement of n^{th} atom is na i.e. $x = na$. Thus, equation (iii) will become:-

$$U_n = U_0 e^{i(\omega t - kna)} \rightarrow (iv)$$

Similarly, the harmonic solution for $(n-1)^{\text{th}}$ atom from its mean position is

$$U_{n-1} = U_0 e^{i(\omega t - k(n-1)a)} = U_0 e^{i(\omega t - kna + ka)}$$

$$U_{n-1} = U_0 e^{i(\omega t - kna)} \cdot e^{ika} \rightarrow (v)$$

For

be same:-

$$U_n = U_0 e^{i(\omega t - kna)} \rightarrow (iv)$$

Similarly: the harmonic solution for $(n-1)^{th}$ atom from its mean position is

$$U_{n-1} = U_0 e^{i(\omega t - k(n-1)a)} = U_0 e^{i(\omega t - kna + ka)}$$

$$U_{n-1} = U_0 e^{i(\omega t - kna)} \cdot e^{ika} \rightarrow (v)$$

For $(n+1)^{th}$ atom the harmonic solution will be:

$$U_{n+1} = U_0 e^{i(\omega t - k(n+1)a)}$$

$$= U_0 e^{i(\omega t - kna - ka)}$$

$$U_{n+1} = U_0 e^{i(\omega t - kna)} \cdot e^{-ika} \rightarrow (vii)$$

Now diff equ (i) w.r.t "t"

$$\frac{dU_n}{dt} = U_0 e^{i(\omega t - kna)} \cdot i\omega$$

$$= i\omega U_0 e^{i(\omega t - kna)}$$

Again differentiate with respect to "t"

$$\frac{d^2 U_n}{dt^2} = (i\omega) U_0 e^{i(\omega t - kna)} \cdot i\omega$$

$$= -\omega^2 U_0 e^{i(\omega t - kna)} \rightarrow (viii)$$

Now putting equation (vii), (vi), (v) in equation (iii)

$$m [-\omega^2 U_0 e^{i(\omega t - kna)}] = \beta [U_0 e^{i(\omega t - kna)} \cdot e^{ika} - 2U_0 e^{i(\omega t - kna)} + U_0 e^{i(\omega t - kna)} \cdot e^{-ika}]$$

$$-m\omega^2 U_0 e^{i(\omega t - kna)} = \beta U_0 e^{i(\omega t - kna)} [e^{ika} - 2 + e^{-ika}]$$

$$-m\omega^2 = \beta [e^{ika} - 2 + e^{-ika}]$$

$$-m\omega^2 = \beta [2 - (e^{ika} + e^{-ika})]$$

$$m\omega^2 = \beta [2 - (e^{ika} + e^{-ika})]$$

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}$$

$$m\omega^2 = \beta [2 - 2 \left(\frac{e^{ika} + e^{-ika}}{2} \right)]$$

$$m\omega^2 = \beta [2 - 2 \cos(ka)] = 2\beta [1 - \cos(ka)]$$

$$m\omega^2 = 2\beta [1 - \cos(ka)]$$

Using half angle formula:

$$1 - \cos \theta = 2 \sin^2 \left(\frac{\theta}{2} \right)$$

$$m\omega^2 = 2\beta [2 \sin^2 \left(\frac{ka}{2} \right)] = 4\beta \sin^2 \left(\frac{ka}{2} \right)$$

$$m\omega = \beta [2 - 2 \cos(ka)] = 2\beta [1 - \cos(ka)]$$

$$m\omega^2 = 2\beta [1 - \cos(ka)]$$

Using half angle formula:

$$1 - \cos\theta = 2 \sin^2\left(\frac{\theta}{2}\right)$$

$$m\omega^2 = 2\beta \left[2 \sin^2\left(\frac{ka}{2}\right) \right] = 4\beta \sin^2\left(\frac{ka}{2}\right)$$



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$$\omega^2 = \frac{4\beta}{m} \sin^2\left(\frac{ka}{2}\right)$$

Taking square root on both sides: we get

$$\omega = \pm 2 \sqrt{\frac{\beta}{m}} \sin\left(\frac{ka}{2}\right)$$

Since angular frequency can never be negative. Thus,

$$\omega = 2 \sqrt{\frac{\beta}{m}} \sin\left(\frac{ka}{2}\right) \rightarrow \text{A}$$

The relation shown in eqn (A) i.e b/w " ω " and " k " is called dispersion relation.

The dispersion relation is defined as:

"In lattice vibrations, dispersion describes the relationship between the frequency of phonon and its wave vector."

Now we will study this dispersion relation at smaller wavelengths, larger wavelengths and also at maximum angular frequency.

Case-I

For longer wavelength ($\lambda \rightarrow \infty$)

When $\lambda \rightarrow \infty$ then $k = \frac{2\pi}{\lambda} \rightarrow 0$. It

It shows medium is dispersive in nature.

Case - III

(Maximum Angular frequency $\omega_{\max}$)

The dispersion relation between ω and k is given as:-

$$\omega = 2 \sqrt{\frac{\beta}{m}} \sin\left(\frac{ka}{2}\right)$$

Maximize angular frequency $\omega_{\max}$

$$\omega_{\max} = 2 \sqrt{\frac{\beta}{m}} \left[\sin\left(\frac{ka}{2}\right) \right]_{\max} = 2 \sqrt{\frac{\beta}{m}} (1)$$

$$\omega_{\max} = 2 \sqrt{\frac{\beta}{m}}$$

Where

$$\left[\sin\left(\frac{ka}{2}\right) \right]_{\max} = 1 = \left[\sin\left(\frac{\pi}{2}\right) \right]_{\max}$$

$$\frac{ka}{2} = \frac{\pi}{2}$$

$$\Rightarrow k = \frac{\pi}{a}$$

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Phase Velocity:-

v_p is defined as:-

$$v_p = \frac{\omega}{k}$$

$$v_p = \frac{2}{k} \sqrt{\frac{\beta}{m}} \sin\left(\frac{ka}{2}\right)$$

For maximum ω , put $ka = \pi$

$$v_p = \frac{2}{k} \sqrt{\frac{\beta}{m}} \sin\left(\frac{\pi}{2}\right) = \frac{2}{k} \sqrt{\frac{\beta}{m}} (1) = \frac{2}{k} \sqrt{\frac{\beta}{m}}$$

Now, by multiplying by a and also by dividing

$$v_p = \frac{2a}{ka} \sqrt{\frac{\beta}{m}} = \frac{2a}{\pi} \sqrt{\frac{\beta}{m}}$$

$$v_p = \frac{2v_0}{\pi} \rightarrow (c)$$

$$= a \sqrt{\frac{\beta}{m}}$$

Group Velocity:

The group velocity v_g is:-

$$v_g = \frac{d\omega}{dk}$$

thus, Using $d\omega$ from above ~~for~~ equation (d)

$$v_g = \frac{1}{dk} \frac{a\sqrt{\frac{\beta}{m}} \cos\left(\frac{ka}{2}\right) dk}{1}$$

$$v_g = a \sqrt{\frac{\beta}{m}} \cos\left(\frac{ka}{2}\right)$$



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$$\therefore a \sqrt{\frac{\beta}{m}} = \text{constant} = v_0$$

$$v_g = v_0 \cos\left(\frac{ka}{2}\right)$$

For maximum ω ; put $ka = 0$

$$v_g = v_0 \cos\left(\frac{0}{2}\right)$$

$$v_g = v_0$$

$$v_g = 0 \rightarrow (f)$$

It shows from equation (e) and (f) that the group of waves transfer no energy. But the individual wave carry some energy.

Thus; frequencies less than or equal to

$$\omega_{\max} = \frac{2v_0}{a}$$

are passed by the lattice. And the frequencies greater than $\frac{2v_0}{a}$ are blocked by lattice.

In this way; mono-atomic linear lattice behaves as "Low-pass filter".

The curve b/w ω and k for mono-atomic linear lattice is plotted:-



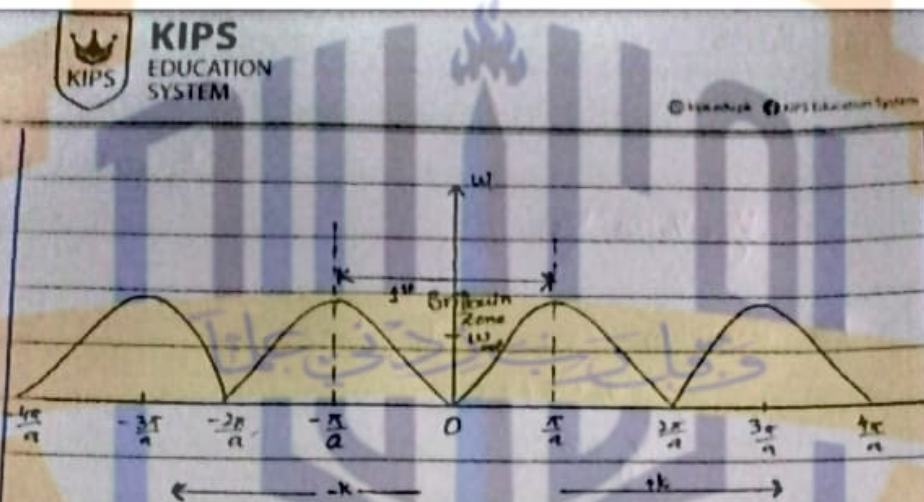
equal to 0

$$\omega_{\max} = \frac{2v_0}{a}$$

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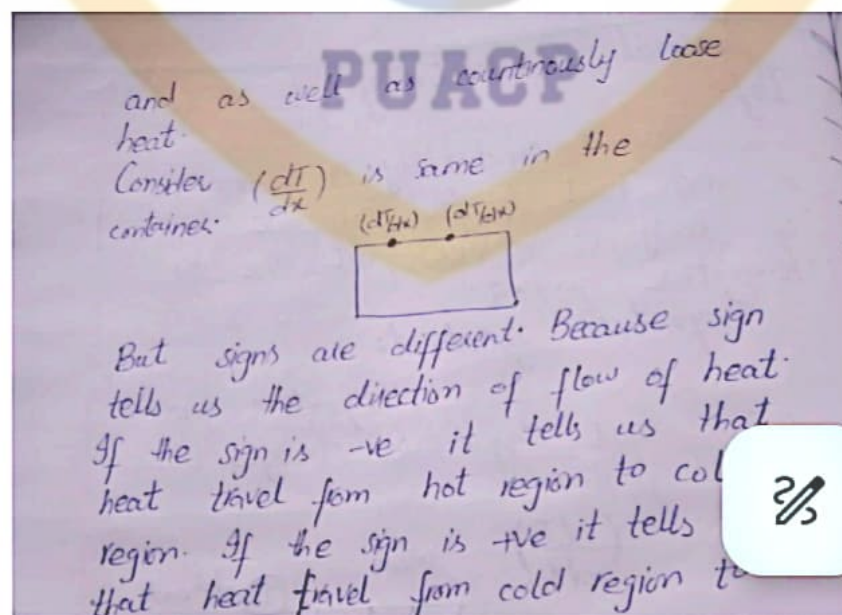
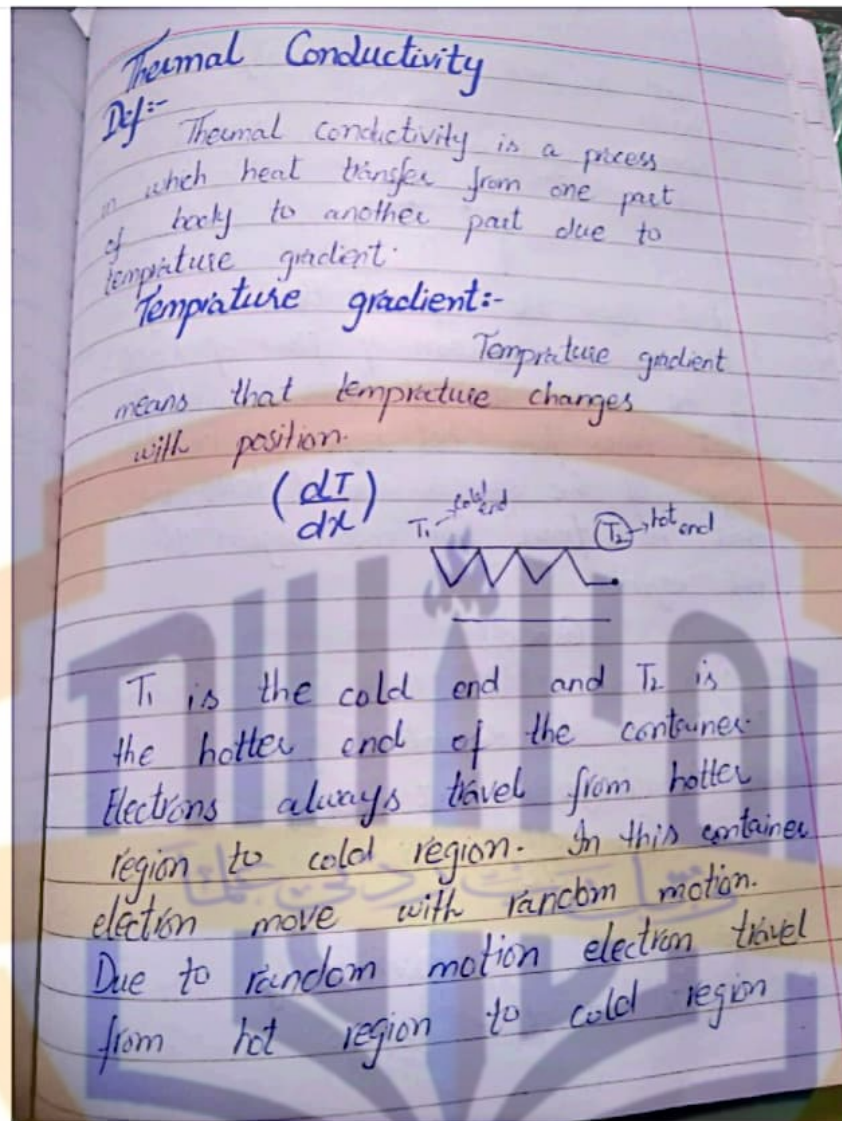


The curve b/w ω and k shown Brillouin Zones.

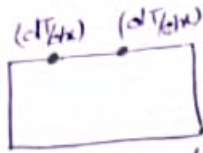
The region $-\frac{\pi}{a} < k < \frac{\pi}{a}$ is called 1st Brillouin Zone.

The range outside the first Brillouin zone but inside $+\frac{2\pi}{a}$ and $-\frac{2\pi}{a}$ is known as 2nd Brillouin zone and so on---.

Thermal conductivity part:



heat.
Consider $\left(\frac{dT}{dx}\right)$ is same in the
containers.



But signs are different. Because sign tells us the direction of flow of heat. If the sign is -ve it tells us that heat travel from hot region to cold region. If the sign is +ve it tells us that heat travel from cold region to hot region.

$$Q \propto \frac{dT}{dx}$$

Means more the temprature gradient more the change in heat energy.

$$Q = K \frac{dT}{dx} \quad \text{--- (1)}$$

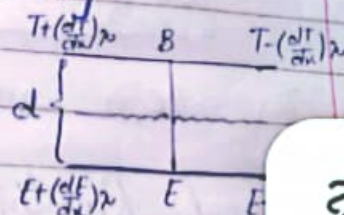
Where K is a proportionality constant. It has two names

Thermal conductivity and coefficient of heat.
Carriers of heat energy:-

Conductors:-
In conductor's heat carriers are electrons.

Insulators:-
In insulators heat carriers are phonons.

Total thermal conductivity:-

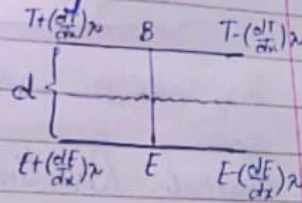


$$\text{no. of electrons} = \frac{1}{6} n v$$



In insulators heat carriers are phonons.

Total Thermal Conductivity:-



no. of electrons = $\frac{1}{6} nv$

flux energy through B to right $Q_{\text{right}} = \frac{1}{6} nv \left(\frac{dE}{dx} \right) \lambda$

flux energy through B to left $Q_{\text{left}} = -\frac{1}{6} nv \left(\frac{dE}{dx} \right) \lambda$

$$Q = Q_{\text{right}} - Q_{\text{left}}$$

$$= \frac{1}{6} nv \lambda \frac{dE}{dx} + \frac{1}{6} nv \lambda \left(\frac{dE}{dx} \right)$$

$$Q = \frac{nv \lambda}{3} \frac{dE}{dx} \quad \text{--- (2)}$$

Apply chain rule:

$$\frac{dE}{dx} = \left(\frac{dE}{dt} \right) \left(\frac{dt}{dx} \right)$$

$$= Cv \frac{dt}{dx}$$

Put the value in eq (2)

$$Q = \frac{nv \lambda}{3} \left(\frac{dE}{dt} \right) \quad \text{--- (3)}$$

Now compare eq (1) & (3)

$$\boxed{K = \frac{1}{3} nv \lambda Cv}$$

This value is called thermal conductivity.

Thermal Resistivity:-

Def:-

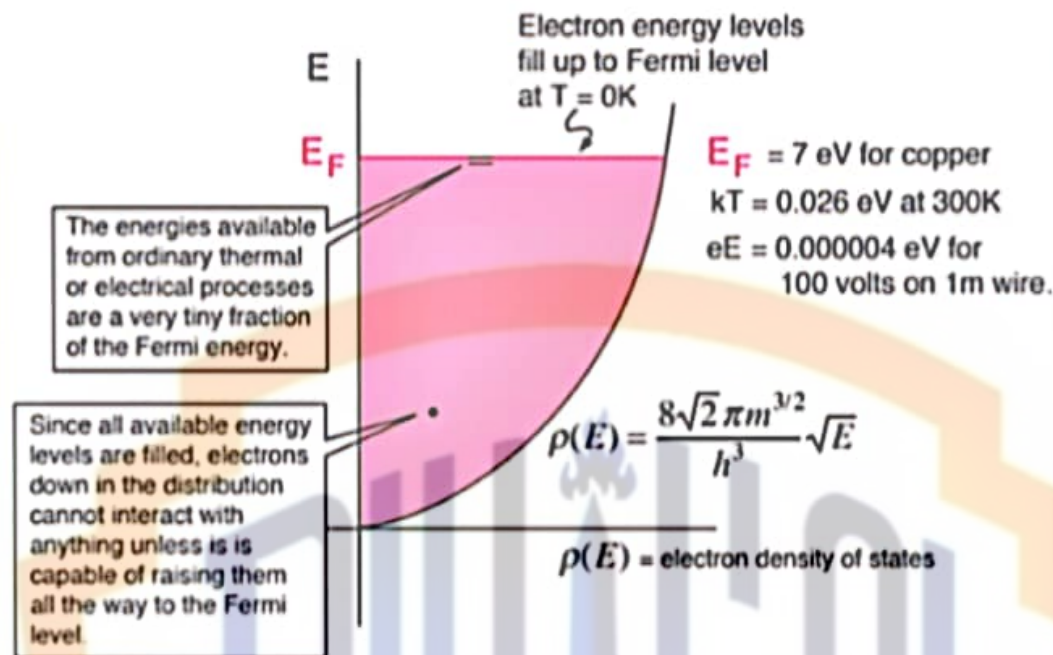
Resistance per unit area is called resistivity.

When two electrons interact with each



Electron Contribution to Specific Heat PART (2)

One of the great mysteries in physics in the early part of the 20th century was why electrons didn't appear to contribute to [specific heat](#). How could they contribute to electrical conduction and heat conduction and not to specific heat?



The electrons in the metal which contribute to conduction are very close to the Fermi level, "[ripples on the Fermi sea](#)". But to contribute to bulk specific heat, all the valence electrons would have to receive energy from the nominal [thermal energy](#) kT . But the [Fermi energy](#) is much greater than kT and the overwhelming majority of the electrons cannot receive such energy since there are no available energy levels within kT of their energy.

The small fraction of electrons which are within kT of the Fermi level does contribute a small [specific heat](#), and this [electron specific heat](#) becomes significant at very low temperatures.

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Electron Specific Heat

Using [Fermi-Dirac statistics](#) (as opposed to [Einstein-Bose](#) for phonons), a small fraction of the electrons are available to participate in specific heat. This fraction contributes a specific heat:

$$C_{\text{electrons}} = \frac{\pi^2 N_A k^2 T}{2E_F} \text{ mole}^{-1}$$

It is important only for low temperatures in metals where it becomes larger than the T -cubed dependent phonon contribution. At those low temperatures it becomes part of the [Einstein-Debye specific heat](#).

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In solids, the contribution of electrons to the specific heat capacity depends on the material's electronic structure and temperature range. In metals, electrons contribute significantly to specific heat capacity due to their mobility and ability to carry thermal energy. This contribution is described by the electronic specific heat term in the electronic theory of metals. In insulators and semiconductors, electrons also contribute, but their effects are often less pronounced compared to metals. Overall, the contribution of electrons to the specific heat capacity of a solid varies depending on the material's properties and temperature conditions.